

The Utility Problem

A well-maintained equipment catalogue — a reference library of the generic asset types a utility installs and operates, complete with their technical specifications — is a foundational asset management artifact. Yet in practice, most electric utilities either lack such a catalogue entirely, or maintain one that is fragmented, outdated, and inconsistent across departments. Equipment specifications may exist in procurement spreadsheets, engineering drawings, manufacturer datasheets, and GIS attribute tables simultaneously, but never in a single authoritative, queryable source.

The absence of a unified equipment catalogue creates cascading problems across the organization. Procurement teams cannot easily determine which transformer models are already in use at the utility, what ratings are standard, or which manufacturers have been approved — leading to non-standardized purchases that complicate spare parts management and increase lifecycle costs. Engineering teams designing new feeders or planning upgrades must hunt for equipment specifications across multiple disconnected systems, often defaulting to outdated or inconsistent data. Asset management teams struggle to perform meaningful fleet-level analysis — such as identifying all transformers of a particular make and vintage that may be at end-of-life — because equipment types are not consistently coded in the GIS or work management system.

The root cause is typically that utilities have never systematically extracted and curated the equipment type information that is already implicitly embedded in their operational data. The installed base of assets, as recorded in the GIS and AMI systems, contains a wealth of equipment specification data — but it is buried in free-text fields, inconsistent naming conventions, and duplicated records. Turning this raw asset data into a clean, structured equipment catalogue requires both data integration capability and intelligent data analysis — something that has historically been too labor-intensive to undertake.

The Data Solutions and Methods

The Awesense Platform aggregates asset attribute data from the utility's GIS, OMS, work order management system, and AMI — normalizing it into a unified asset data model (Awesense Energy Data Model - EDM) that brings together every recorded instance of every installed asset. This installed base dataset forms the raw material from which an equipment catalogue can be systematically derived and curated.

The catalogue creation process applies a combination of data analysis techniques to the installed base. First, attribute clustering and deduplication algorithms are applied to the population of recorded assets to identify the distinct equipment types present in the fleet. For example, across thousands of distribution transformer records, the algorithm groups records by their combination of key attributes — rated kVA, primary voltage, secondary voltage, insulation type, phase configuration, and manufacturer — and resolves minor naming inconsistencies and free-text variations to produce a clean list of unique transformer types actually in service. The

Equipment and Asset Type Catalogue

[UC55] - Awesense Platform Use Case



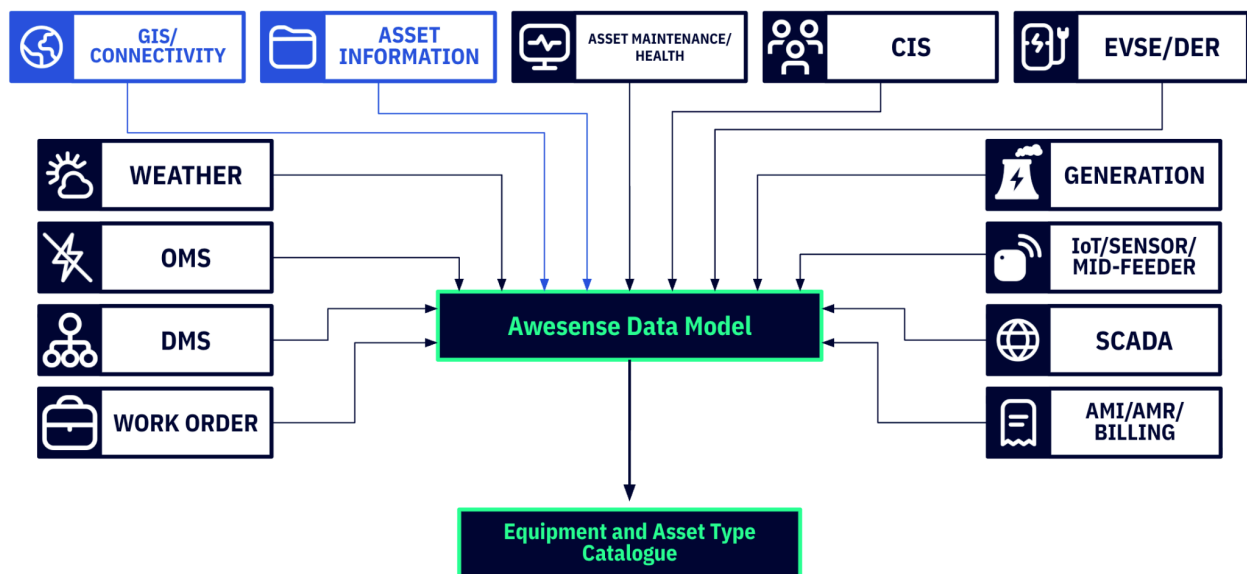
same approach is applied to cables, switches, fuses, reclosers, and other primary distribution equipment classes.

Second, statistical analysis of the resulting clusters provides enriched catalogue entries. For each identified equipment type, the platform computes the count of installed units, the geographic and feeder distribution across the service territory, the age distribution based on installation dates, and the frequency of association with maintenance events or outage records. This contextual data transforms a bare specification list into an operationally informed catalogue — one that tells engineers not just what types of equipment exist, but how many are deployed, where they are concentrated, and how they have historically performed.

The resulting catalogue is maintained as a living dataset within the platform. As new GIS imports are ingested, the platform detects newly installed equipment types and flags them for review, ensuring the catalogue remains current without manual effort. Users can query, filter, and export the catalogue by asset class, rating, manufacturer, or any combination of technical attributes — directly supporting procurement standardization, engineering design reference, and fleet-level asset management decisions.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.